Tennis Analysis Pipeline

This project provides an automated pipeline for analyzing tennis matches using deep learning models to track ball movement, detect court boundaries, and identify bounce events.

Usage

To run the analysis on a video file, use the following command:

```
python3 main.py \
  --path_ball_track_model tracknet_model.pt \
  --path_court_model court_model.pt \
  --path_bounce_model bounce_model.cbm \
  --path_input_video tennis_video.mp4 \
  --path_output_video output.mp4
```

Arguments

Argument	Description
`--path_ball_track_model`	Path to the pre-trained TrackNet model (PyTorch .pt) for ball tracking.
`--path_court_model`	Path to the court detection model (.pt) to identify court lines.
`--path_bounce_model`	Path to the CatBoost (.cbm) model used for bounce event detection.
`--path_input_video`	The source MP4 video file of the tennis match.
`--path_output_video`	The filename/path where the processed video will be saved.

System Requirements

- Python 3.x
- PyTorch (for .pt models)
- CatBoost (for .cbm models)
- OpenCV (for video processing)